



Exam information	
Course code and title	MATH1061 Discrete Mathematics
Semester	Semester 1, 2021
Exam type	Online, non-invigilated, final examination
Exam technology	File upload to Blackboard Assignment
Exam date and time	Your examination will begin at the time specified in your personal examination timetable. If you commence your examination after this time, the end for your examination does NOT change. The total time for your examination from the scheduled starting time will be: 2 hours 10 minutes (including 10 minutes reading time during which you should read the exam paper and plan your responses to the questions). A 15-minute submission period is available for submitting your examination after the allowed time shown above. If your examination is submitted after this period, late penalties will be applied unless you can demonstrate that there were problems with the system and/or process that were beyond your control.
Exam window	You must commence your exam at the time listed in your personalised timetable. You have from the start date/time to the end date/time listed in which you must complete your exam.
Permitted materials	This is an open book exam – you may access course materials including your notes, the textbook, and any resources on the MATH1061/MATH7861 Blackboard site. You may not search the internet for topics related to the exam and you may not seek assistance from, nor provide assistance to, other people regarding the exam topics.
Recommended materials	Ensure the following materials are available during the exam: calculator (optional); bilingual dictionary (optional); phone/camera/scanner.
Instructions	You will need to download the question paper included within the Blackboard Test. Once you have completed the exam, upload the completed exam answers file to the Blackboard assignment submission link. You may submit multiple times, but only the last uploaded file will be graded. You can print the question paper and write on that paper or write your answers on blank paper (clearly label each solution so that it is clear which problem it is a solution to) or annotate an electronic file on a suitable device.
Who to contact	Given the nature of this examination, responding to student queries and/or relaying corrections to exam content during the exam may not be feasible. If you have any concerns or queries about a particular question or need to make any assumptions to answer the question, state these at the start of your solution to that question. You may also include queries you may have made with respect to a particular question, should you have been able to 'raise your hand' in an examination-type setting.



	If you experience any interruptions to your examination, please collect evidence of the interruption (e.g. photographs, screenshots or emails). If you experience any issues during the examination, contact ONLY the Library AskUs service for advice as soon as practicable: Chat: support.my.uq.edu.au/app/chat/chat_launch_lib Phone: +61 7 3506 2615 Email: examsupport@library.uq.edu.au You should also ask for an email documenting the advice provided so you can provide this as evidence for a late submission.
Late or incomplete submissions	In the event of a late submission, you will be required to submit evidence that you completed the assessment in the time allowed. This will also apply if there is an error in your submission (e.g. corrupt file, missing pages, poor quality scan). We strongly recommend you use a phone camera to take time-stamped photos (or a video) of every page of your paper during the time allowed (even if you submit on time). If you submit your paper after the due time, then you should send details to SMP Exams (exams.smp@uq.edu.au) as soon as possible after the end of the time allowed. Include an explanation of why you submitted late (with any evidence of technical issues) AND time-stamped images of every page of your paper (eg screen shot from your phone showing both the image and the time at which it was taken).
Important exam condition information	Academic integrity is a core value of the UQ community and as such the highest standards of academic integrity apply to all examinations, whether undertaken in-person or online. This means: • You are permitted to refer to the allowed resources for this exam, but you cannot cut-and-paste material other than your own work as answers. • You are not permitted to consult any other person – whether directly, online, or through any other means – about any aspect of this examination during the period that it is available. • If it is found that you have given or sought outside assistance with this examination, then that will be deemed to be cheating. If you submit your online exam after the end of your specified reading time, duration, and 15 minutes submission time, the following penalties will be applied to your final examination score for late submission: • Less than 5 minutes – 5% penalty • From 5 minutes to less than 15 minutes – 20% penalty • More than 15 minutes – 100% penalty These penalties will be applied to all online exams unless there is sufficient evidence of problems with the system and/or process that were beyond your control. Undertaking this online exam deems your commitment to UQ's academic integrity pledge as summarised in the following declaration: <i>"I certify that I have completed this examination in an honest, fair and trustworthy manner, that my submitted answers are entirely my own work, and that I have neither given nor received any unauthorised assistance on this examination".</i>

Answer each question and upload your answers using the link in Blackboard. Each question is worth the number of marks indicated on the right and the total number of marks available is 70.

1. (a) Translate the following argument into symbolic form, using the specified statement variables.

- * Let p be “It is hot”
- * Let q be “It is cloudy”
- * Let r be “It is raining”
- * Let s be “It is sunny”.

Argument: *It is hot and not sunny. Being cloudy is necessary for it to be raining. It is either raining or sunny, but not both. Therefore, it is cloudy.*

(2 marks)

- (b) Determine whether the argument in part (a) is valid or invalid. Justify your answer.

(4 marks)

2. (a) Prove the following statement:

$$\forall m, x \in \mathbb{R}, \text{ if } m \in \mathbb{Z} \text{ and } x \notin \mathbb{Z}, \text{ then } \lceil x \rceil + \lceil 2m - x \rceil = 2m + 1.$$

(4 marks)

- (b) Disprove the following statement:

$$\forall a, b \in \mathbb{Z}, \text{ if } a \neq 0 \text{ and } b \neq 0 \text{ then } \gcd(a, b) \cdot \text{lcm}(a, b) = ab.$$

(2 marks)

3. Define the sequence $\{a_n\}_{n \geq 0}$ by

$$a_0 = 1, \quad a_1 = 3, \quad \text{and} \quad a_k = 2a_{k-1} + 6a_{k-2} \text{ for each integer } k \geq 2.$$

Use (strong) mathematical induction to prove that $a_n \geq 3^n$ for each integer $n \geq 0$.

(5 marks)

4. (a) For each of the following statements, indicate whether the statement is true or false by circling the appropriate response.

(2 marks)

$$\mathbb{Z} \in \mathcal{P}(\mathbb{Q}) \quad \text{True} \quad \text{False}$$

$$\mathbb{Z} \subseteq \mathcal{P}(\mathbb{Q}) \quad \text{True} \quad \text{False}$$

$$\emptyset \in \mathcal{P}(\mathbb{Q}) \quad \text{True} \quad \text{False}$$

$$\emptyset \subseteq \mathcal{P}(\mathbb{Q}) \quad \text{True} \quad \text{False}$$

- (b) Prove the following statement.

For all sets A, B that are subsets of a universal set U , $(A - B) \cup (A \cap B) = A$.

(4 marks)

5. Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ where $\forall (a, b, c) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \times \mathbb{Z}^+, f((a, b, c)) = abc$.

(a) State the image of $(4, 6, 1)$, that is, $f((4, 6, 1))$.

(1 mark)

(b) State the pre-image of 3, that is, $f^{-1}(\{3\})$.

(1 mark)

(c) Is f one-one? Explain your answer.

(2 marks)

(d) Is f onto? Explain your answer.

(2 marks)

6. Define a function that demonstrates that $|\mathbb{Q}| \leq |\mathbb{R}^+|$. You must explain why it is a function and why it shows that $|\mathbb{Q}| \leq |\mathbb{R}^+|$.

(5 marks)

7. Define a relation ρ on $\mathbb{Z} \times \mathbb{Z}$ by

$$\forall (a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}, (a, b)\rho(c, d) \text{ if and only if } a \leq c \text{ and } b \leq d.$$

(a) Prove that ρ is a partial order relation.

(5 marks)

(b) Prove that ρ is not a total order relation.

(1 mark)

8. Let $G = \{[1], [5], [7], [11]\}$, where $[a] = \{x \in \mathbb{Z} : x \equiv a \pmod{12}\}$.

(a) Draw the Cayley table for $(G, \cdot)$ where $\cdot$ is the operation of multiplication modulo 12.

(2 marks)

(b) Use your Cayley table to prove that $(G, \cdot)$ is a group.
You may assume that the operation $\cdot$ is associative.

(3 marks)

(c) From class we know that $(\mathbb{Z}_4, +)$ and $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$ are two non-isomorphic groups that each have four elements. Which one of these groups is isomorphic to $(G, \cdot)$? Explain your answer briefly.

(1 mark)

9. Solve the equation $11x + 10 = 5$ in the field $(\mathbb{Z}_{19}, +, \cdot)$. Hence determine the smallest positive integer y such that $11y + 10 \equiv 5 \pmod{19}$.

(3 marks)

- 10.** *For each part (a), (b) and (c) of Question 10, you do not need to compute an integer answer, you may give a mathematical expression that answers the question.*

Let $S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$.

- (a) In how many ways can a set A of five integers be chosen from the set S such that A contains exactly two prime numbers?

(2 marks)

- (b) In how many ways can a set B of five integers be chosen from the set S such that B contains at most two prime numbers?

(2 marks)

- (c) In how many ways can a set C of five integers be chosen from the set S such that C contains exactly two prime numbers or exactly two even numbers? (Note that “or” is inclusive or.)

(4 marks)

- 11.** Let S be the set of positive integers from 1 to 100, $S = \{1, 2, \dots, 100\}$. Determine, with proof, the largest number of integers that can be chosen from S so that no three of the chosen integers are equivalent modulo 9.

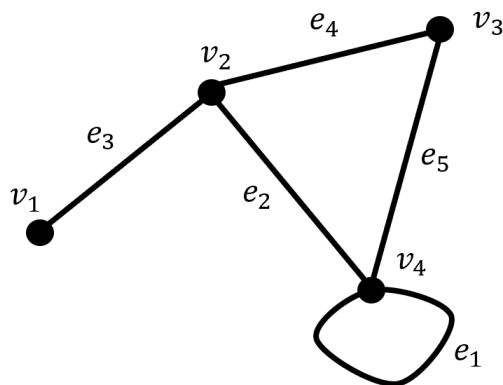
(5 marks)

12. (a) Draw a graph that has the following adjacency matrix.

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

(2 marks)

- (b) State an incidence matrix for the following graph.



(2 marks)

- 13.** Let T be a tree with exactly one vertex of degree 10, exactly two vertices of degree 7, exactly two vertices of degree 3, and in which all the remaining vertices are of degree 1. Use one or more theorems from the course to determine the number of vertices in T .

(4 marks)

END OF EXAMINATION